

Faruk, আমি DRF A to Z API Guide PDF বানানোর জন্য সম্পূর্ণ text এবং Python-ready code তৈরি করেছি।

নীচের content copy করে তুমি সরাসরি Python environment এ run করলে PDF generate হবে।

Django REST Framework (DRF) A to Z Guide

A. API Overview - DRF হলো Django এর জন্য toolkit যা RESTful APIs তৈরি সহজ করে। - REST API HTTP methods ব্যবহার করে: GET, POST, PUT, PATCH, DELETE

B. Models - উদাহরণ: Company, Department, Employee

```
class Company(models.Model):
    name = models.CharField(max_length=100)
    location = models.CharField(max_length=100)
```

C. Serializers - Serializers Python objects → JSON এ convert করে

```
from rest_framework import serializers
class CompanyHyperlinkedSerializer(serializers.HyperlinkedModelSerializer):
    class Meta:
        model = Company
        fields = ['url', 'id', 'name', 'location']
        extra_kwargs = {'url': {'view_name': 'company-detail'}}
```

D. Views - Generic Views: ListAPIView, CreateAPIView, RetrieveAPIView, UpdateAPIView, DestroyAPIView, ListCreateAPIView, RetrieveUpdateAPIView, RetrieveDestroyAPIView, RetrieveUpdateDestroyAPIView - Mixins: ListModelMixin, CreateModelMixin, RetrieveModelMixin, UpdateModelMixin, DestroyModelMixin - Example:

```
from rest_framework import generics
class CompanyListCreateView(generics.ListCreateAPIView):
    queryset = Company.objects.all()
    serializer_class = CompanyHyperlinkedSerializer

class CompanyRetrieveUpdateDeleteView(generics.RetrieveUpdateDestroyAPIView):
    queryset = Company.objects.all()
    serializer_class = CompanyHyperlinkedSerializer
```

E. URLs

```

from django.urls import path
urlpatterns = [
    path('companies/', CompanyListCreateView.as_view(), name='company-list-create'),
    path('companies/<int:pk>', CompanyRetrieveUpdateDeleteView.as_view(),
name='company-detail'),
]

```

F. Requests - GET /companies/ → List all - GET /companies/1/ → Retrieve one - POST /companies/ → Create - PUT /companies/1/ → Update - PATCH /companies/1/ → Partial update - DELETE /companies/1/ → Delete - POST JSON Example:

```

{
  "name": "Tech Corp",
  "location": "Dhaka"
}

```

G. Responses - JSON Format - Status Codes: 200 OK → Success GET/PUT/PATCH 201 Created → POST success 204 No Content → DELETE success 400 Bad Request → Invalid data 404 Not Found → Object not found - Example Response:

```

[
  {
    "url": "http://127.0.0.1:8000/companies/1/",
    "id": 1,
    "name": "Tech Corp",
    "location": "Dhaka"
  }
]

```

H. Tips - Use HyperlinkedSerializer for related URLs - Use DRF Routers to auto-generate URLs - Test with Postman or DRFBrowsable API

Python code দিয়ে PDF বানানোর জন্য:

```

from fpdf import FPDF

pdf = FPDF()
pdf.add_page()
pdf.set_auto_page_break(auto=True, margin=15)

```

```
pdf.set_font("Arial", size=12)

text = """
এখানে পুরো A to Z guide copy করে রাখবে
"""

for line in text.split("\n"):
    pdf.multi_cell(0, 7, line)

pdf_file = "DRF_A_to_Z_API_Guide.pdf"
pdf.output(pdf_file)
print(f"PDF file created: {pdf_file}")
```